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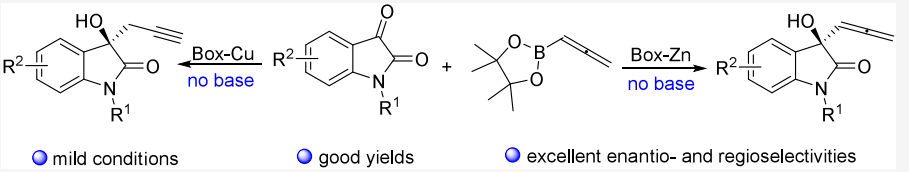
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Enantioselective Propargylation and Allenylation of Isatins Using Chiral Box-Copper/Zinc Catalysts

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ABSTRACT: A method for the asymmetric propargylation and allenylation of isatins catalyzed using chiral Box-copper/zinc catalysts is described. By modulation of the metal and solvent, a series of chiral 3-hydroxy-3-propargyl-2-oxindoles and 3-hydroxy-3-allenyl-2-oxindoles are selectively synthesized in good yields with excellent enantio- and regioselectivities.

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Chiral 3-hydroxy-2-oxindoles are the core structural units of numerous biologically active natural products and drug intermediates (Figure 1). For instance, convolutamydine

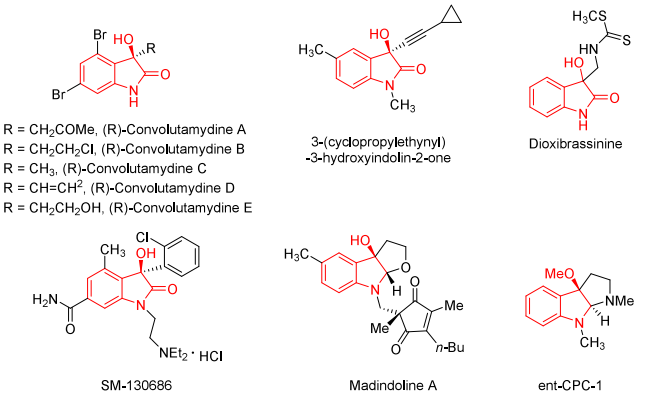


Figure 1. Representative biologically active products synthesized via chiral 3-hydroxy-2-oxindoles.

A−E exhibit potential antinociceptive[1] and anticancer activity,[2] 3-(cyclopropylethynyl)-3-hydroxyindolin-2-one acts as an HIV-1 reverse transcriptase inhibitor,[3] dioxibrassinine serves as a phytoalexin with antifungal activity,[4] SM-130688 is a novel orally active growth hormone secretagogue,[5] madindoline A acts as an antiosteoporosis agent,[6] and ent-CPC-1 is a newly discovered tryptamine-derived dimeric alkaloid from the seeds and rinds of *Chimonanthus praecox f.* concolor.[7] On the other hand, 3-hydroxy-3-propargyl-2-oxindoles and 3-hydroxy-3-allenyl-2-oxindoles are synthons of many important structures,[8] particularly for synthesizing diverse substituted 2,5-dihydrofurans.[9] Consequently, these compounds have garnered significant attention from synthetic chemists, and asymmetric synthesis of chiral indole and oxindole derivatives remains a focal point of current research.[10]

The Barbier-type reaction is one of the most common methods for forming carbon−carbon bonds between propargyl bromide and isatins through the addition of different metals (such as Zn, Sn, In, etc.). To selectively obtain propargyl or allenyl alcohols, different metals are required in a Barbier-type reaction, because propargylmetals can undergo a metallotropic rearrangement (Scheme 1A).[11]

To our knowledge, asymmetric syntheses of 3-hydroxy-3-propargyl-2-oxindoles and 3-hydroxy-3-allenyl-2-oxindoles are relatively scarce. In terms of organocatalysis, Hoveyda and co-workers catalyzed the asymmetric addition reaction between isatins and allenylboronic acid pinacol ester to achieve 3-hydroxy-3-allenyl-2-oxindoles using simple organic molecule **1A**,[12] and chiral 3,3′-Br_2−BINOL **1B** was proved to be an effective catalyst to synthesize chiral 3-hydroxy-3-propargyl-2-oxindoles (Scheme 1B).[13] In terms of metal catalysis, a complex of (S)-SEGPHOS **1C** and copper(II) triflate effectively catalyzes the reaction between isatins and allenylboronic acid pinacol ester to afford chiral 3-hydroxy-3-propargyl-2-oxindoles with moderate to good enantioselectivity.[14] Conversely, chiral 3-hydroxy-3-allenyl-2-oxindoles are obtained using a complex of proline-derived ligand **1D** and zinc triflate (Scheme 1C).[15]

Herein, we successfully achieved the enantioselective propargylation and allenylation of isatins using commercially

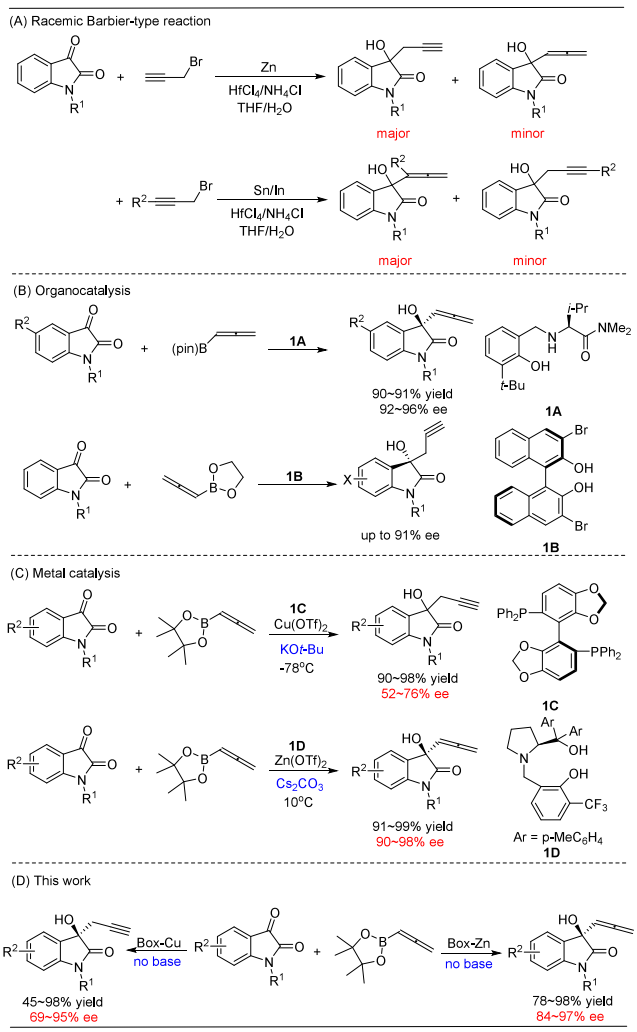
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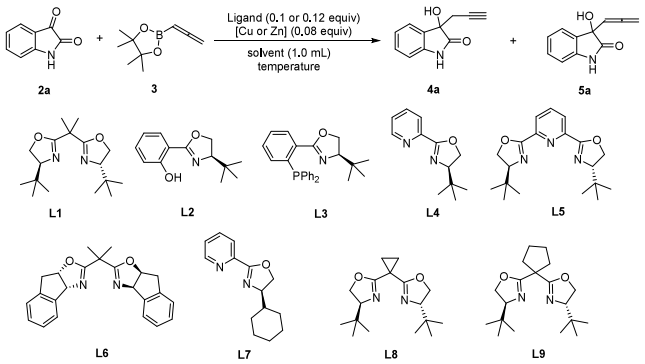
Scheme 1. Catalytic Propargylation and Allenylation of Isatins



available bisoxazoline ligands. By simply adjusting the metal catalyst and solvent, we selectively obtained 3-hydroxy-3-propargyl-2-oxindoles and 3-hydroxy-3-allenyl-2-oxindoles with good yields, excellent enantio- and regioselectivities, without the need for a base (Scheme 1D).

We initiated our investigation by propargylication of isatin with allenylborate using the L−Cu complex. First, we examined six different commercially available ligands **L1**−**L6** using $Cu(MeCN)_4BF_4$ as a copper source in DCM and found that bisoxazoline ligand **L6** was the most efficient ligand, which can selectively synthesize propargyl alcohol **4a** (Table 1, entries 1−6). Subsequently, various copper salts were screened, and the experimental results indicated poor enantioselectivity when $Cu(COOCH_3)_2$, $CuBr_2$ and CuBr were used (entries 8−9 and 12). Although $Cu(SO_3CF_3)_2$ yielded products with good regio- and enantioselectivities, the conversion rate remained unsatisfactory (entry 7). $Cu(MeCN)_4SO_3CF_3$ was identified as the optimal copper source, yielding excellent results in terms of both yield and selectivity (entry 10). The use of isopropyl alcohol (IPA) as a solvent led to a slight improvement in conversion and enantioselectivity (Supporting Information). Surprisingly, allenyl alcohol was afforded as the major product when copper salt is replaced with zinc salt. Subsequently, we optimized the conditions for the zinc-catalyzed allenyl alcohol

Table 1. Optimization of the Reaction Conditions[a]



Entry	Ligand/[Cu or Zn]	4a:5a[b]	Conversion (%)[c]	ee (%)
1	**L1**/$Cu(MeCN)_4BF_4$	96:4	38	4
2	**L2**/$Cu(MeCN)_4BF_4$	94:6	66	5
3	**L3**/$Cu(MeCN)_4BF_4$	92:8	74	16
4	**L4**/$Cu(MeCN)_4BF_4$	97:3	96	64
5	**L5**/$Cu(MeCN)_4BF_4$	92:8	62	34
6	**L6**/$Cu(MeCN)_4BF_4$	97:3	56	89
7	**L6**/$Cu(SO_3CF_3)_2$	99:1	65	90
8	**L6**/$Cu(COOCH_3)_2$	95:5	85	25
9	**L6**/$CuBr_2$	70:30	18	11
10	**L6**/$Cu(MeCN)_4SO_3CF_3$	99:1	97	90
11	**L6**/$Cu(MeCN)_4PF_6$	92:8	68	91
12	**L6**/CuBr	94:6	81	23
13	**L1**/$Zn(BF_4)_2{\cdot}xH_2O$	6:94	100	66
14	**L3**/$Zn(BF_4)_2{\cdot}xH_2O$	9:91	100	15
15	**L4**/$Zn(BF_4)_2{\cdot}xH_2O$	5:95	95	18
16	**L6**/$Zn(BF_4)_2{\cdot}xH_2O$	20:80	100	23
17	**L7**/$Zn(BF_4)_2{\cdot}xH_2O$	30:70	100	1
18	**L8**/$Zn(BF_4)_2{\cdot}xH_2O$	3:97	100	91
19	**L9**/$Zn(BF_4)_2{\cdot}xH_2O$	6:94	100	91
20	**L8**/$Zn(CF_3SO_3)_2$	45:55	100	90
21	**L8**/$Zn(ClO_4)_2{\cdot}6H_2O$	4:96	100	93
22	**L8**/$Zn(CH_3COO)_2$	15:85	100	1
23	**L8**/$ZnCl_2$	4:96	47	3

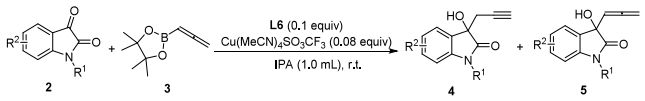
[a]Unless noted otherwise, reactions were performed with **2a** (0.25 mmol, 1.0 equiv), **3** (0.05 mmol, 2.0 equiv), ligand (0.025 or 0.03 mmol, 0.1 or 0.12 equiv), [Cu or Zn] (0.02 mmol, 0.08 equiv) in DCM (1.0 mL) under an argon atmosphere. [b]The ratios of **4a** to **5a** were measured by chiral HPLC analysis. [c]The resulting conversion was measured by chiral HPLC analysis.

formation. Our findings indicated that commercially available bisoxazoline ligand **L8** was the most effective ligand (entries 13−19), while $Zn(ClO_4)_2{\cdot}6H_2O$ served as the superior zinc source (entries 20−23). Detailed optimization of other reaction parameters, including solvents, temperature, ligand, and metal salt loadings, is provided in the Supporting Information. Additionally, the addition of bases such as KO^tBu or Cs_2CO_3 was found to be detrimental to the reaction.

With the optimal reaction conditions in hand, a variety of isatins with different electronic and steric properties were surveyed to study the applied range of the developed methodology (Table 2). This approach proved to be suitable for both protected and unprotected isatins. For unprotected isatins, whatever their electronic nature, the presence of substituents at the C4, C5, C6, and C7 positions on the aromatic ring could be used to obtain propargyl alcohols **4a**−**j** with moderate to excellent yields and enantioselectivities (entries 1−10).

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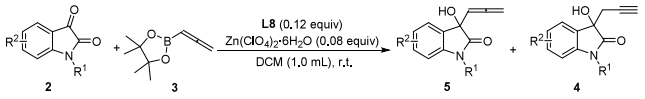
Table 2. Substrate Scope for Enantioselective Propargylation[a]



Entry	R^1	R^2	Product	Yield (%)[b]	ee (%)	Selectivity[c] 4:5
1	H	H	**4a**	98	91	99:1
2	H	4-F	**4b**	97	88	99:1
3	H	5-Cl	**4c**	49	86	>99:1
4	H	5-Me	**4d**	74	90	99:1
5	H	6-Cl	**4e**	45	80	98:2
6	H	7-F	**4f**	97	87	99:1
7	H	7-Cl	**4g**	88	84	99:1
8	H	7-Br	**4h**	86	85	99:1
9	H	7-CF_3	**4i**	86	80	99:1
10	H	7-Me	**4j**	50	72	98:2
11	H	5,6-F	**4k**	93	84	97:3
12	H	4-Cl, 5-F	**4l**	78	69	>99:1
13	1-Ph	H	**4m**	82	92	98:2
14	1-CH_2Ph	H	**4n**	61	91	98:2
15	1-Me	H	**4o**	85	95	>99:1
16	1-Me	5-F	**4p**	89	90	99:1
17	1-Me	5-Br	**4q**	98	87	99:1

[a]Unless noted otherwise, the resulting homopropargyl alcohols was produced under the corresponding optimal conditions. [b]The yield was the isolated yield. [c]The ratios were measured by chiral HPLC analysis.

Table 3. Substrate Scope for Enantioselective Allenylation[a]



Entry	R^1	R^2	Product	Yield (%)[b]	ee (%)	Selectivity[c] 5:4
1	H	H	**5a**	90	93	96:4
2	H	4-F	**5b**	97	84	94:6
3	H	5-Cl	**5c**	94	93	93:7
4	H	5-Me	**5d**	78	92	95:5
5	H	6-Cl	**5e**	83	92	93:7
6	H	7-F	**5f**	97	92	96:4
7	H	7-Cl	**5g**	96	95	95:5
8	H	7-Br	**5h**	98	96	96:4
9	H	7-CF_3	**5i**	96	96	98:2
10	H	7-Me	**5j**	95	92	82:18
11	H	5,7-Br	**5k**	90	96	98:2
12	1-Ac	H	**5l**	94	96	99:1
13	1-Ph	H	**5m**	96	95	98:2
14	1-CH_2Ph	H	**5n**	97	95	97:3
15	1-Me	H	**5o**	97	97	98:2
16	1-Me	5-F	**5p**	97	95	94:6
17	1-Me	5-Br	**5q**	97	96	96:4

[a]Unless noted otherwise, the resulting homopropargyl alcohols was produced under the corresponding optimal conditions. [b]The yield was the isolated yield. [c]The ratios were measured by chiral HPLC analysis.

Notably, electron-withdrawing groups at the C7 position resulted in particularly good yields and ee values (entries 6−9). When 7-methylisatin was used as a substrate, the yield and ee value significantly decreased, which may be due to the electronic properties of the methyl group (entry 10). For disubstituted isatins, 5,6-difluoroisatin gave the target product with a higher yield and ee value, whereas 4-chloro-5-fluoroisatin exhibited the lowest enantioselectivity (69% ee) (entries 11−12). Substitution at the N1 position with methyl, phenyl, or benzyl groups gave better enantioselectivity, and a methyl substituent at N1 position gave the best result with an ee up to 95% (entries 13−15). Similarly, 5-substituted methylisatins also furnished good results (entries 16−17). Fortunately, high regioselectivities were obtained in all cases.

Various substituted isatins were then reacted with allenyl boronate **3** under the optimized conditions (Table 3). Regardless of the presence or absence of a protecting group on the N atom of isatin, the reaction proceeded smoothly to afford the allenyl alcohols **5a−q** in excellent yields and selectivities. Isatins bearing electron-withdrawing groups such as F, Cl, Br, and CF_3 at the C4, C5, C6 and C7 positions delivered allenyl products **5** in high yields with high enantioselectivities and good regioselectivities (entries 2−3, 5−9). However, when methyl groups were present at the C5 and C7 positions of isatin, both the yield and regioselectivity decreased (entries 4 and 10). The reaction was also effective for disubstituted isatin. 5,7-Dibromoisatin yielded the target product **5k** in high yield with excellent enantioselectivity and regioselectivity (entry 11). The presence of a protecting group on the N atom of isatin had a minimal impact on yield and selectivity, as evidenced by the excellent results obtained with 1-acetylisatin, 1-phenylisatin, 1-benzylisatin, and 1-methylisatin (entries 12−15). Gratifyingly, when electron-withdrawing groups (F and Br) were introduced on the 1-methylisatin, the desired product was obtained in 97% yield with 95%−96% ee.

According to previous literature reports,[16] a probable mechanism is illustrated in Scheme 2. Initially, a metal−boron exchange reaction occurs between allenylboronic acid pinacol esters and **IA**/**IB**, forming propargyl spieces **IIA** and **IIB**. Due to the low energy barrier for isomerization between **IIA** and its isomer **IIA′**, rapid isomerization can occur. The addition reaction between **IIA′** and **2a** proceeds more rapidly, leading to the formation of propargyl alcohol product **4a**. In contrast, **IIB** does not undergo significant isomerization and complexes with istain **2a** to form intermediate **IIIB**. Subsequently, the zinc-propargyl complex attack of the carbonyl group isolates **2a** from the *Re*-face. This hypothesis is supported by ^{1}H NMR data (see Figure S1 and Figure S2 in the Supporting Information).

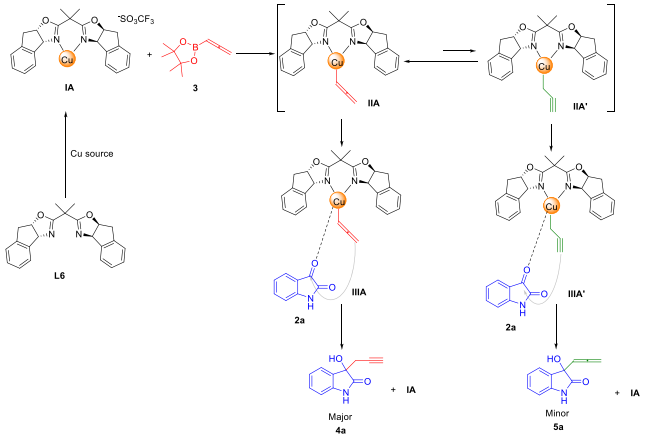
To demonstrate the practical utility of the method, the reaction was conducted on 1 mmol scale (Scheme 3). For the copper-catalyzed propargylization reaction, propargyl alcohol **4a** was afforded with 98% yield and 90% ee (**4a**:**5a** = 99:1). Unfortunately, for zinc-catalyzed allenylation reactions, both the yield and selectivity were reduced (74% yield, 80% ee, **5a**:**4a** = 84:16), which may be attributed to the poor solubility of the reaction intermediates.

In summary, we have developed a mild protocol for the propargylation and allenylation of isatin derivatives using a commercially available organoboron reagent under Box-Cu/Zn catalysis. The reaction conditions are well-tolerated by both protected and unprotected isatins, yielding a series of propargyl and allenyl tertiary alcohols in moderate to high yields with excellent enantio- and regioselectivities. Notably, the ee value of propargyl alcohol reached 95%, surpassing that of the SEGPHOS-Cu(II) catalytic system reported previously. The

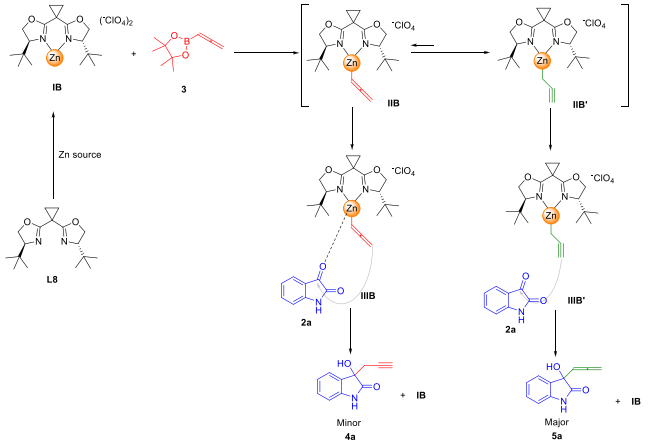
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Scheme 2. Proposed Mechanism

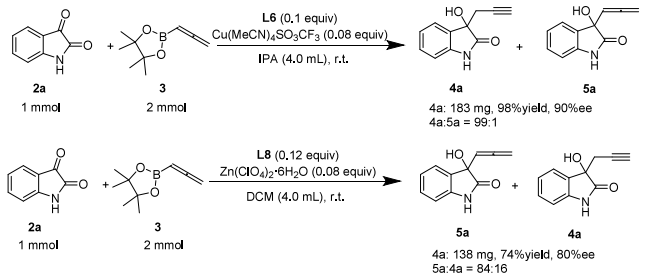
(a) Enantioselective propargylation



(b) Enantioselective allenylation



Scheme 3. Larger Scale Reactions[a,b]



[a]The yield was the isolated yield. [b]The ratios were measured by chiral HPLC analysis.

ee value of allenyl alcohol is comparable to literature reports, but our methodology offers the advantage of using a commercially available ligand without the need for a base.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.4c02850.

Experimental procedures, including compound characterization data, and product spectra (PDF)

FAIR data, including the primary NMR FID files, for compounds **4a**−**4q** and **5a**−**5q** (ZIP)

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Notes

The authors declare no competing financial interest.

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